

EE675

Assignment 1

Q2(a). Value Iteration Function :

Value Iteration is an iterative dynamic programming method used to compute the optimal value function. In this implementation, the value of each state is updated by taking the maximum expected return over all possible actions. Since the exact transition probabilities are not available, a sampling-based approach is used. For each state-action pair, multiple simulations are performed to estimate the expected reward and next state value. The process is repeated until the values stabilize or a maximum number of iterations is reached.

Q2(b). Induced Policy Function :

It is obtained from the Q-function computed using Value Iteration. For each state, the action with the highest Q-value is selected. This results in a deterministic policy that represents the best action to take in each state according to the learned value function.

Q2(c). Policy Iteration :

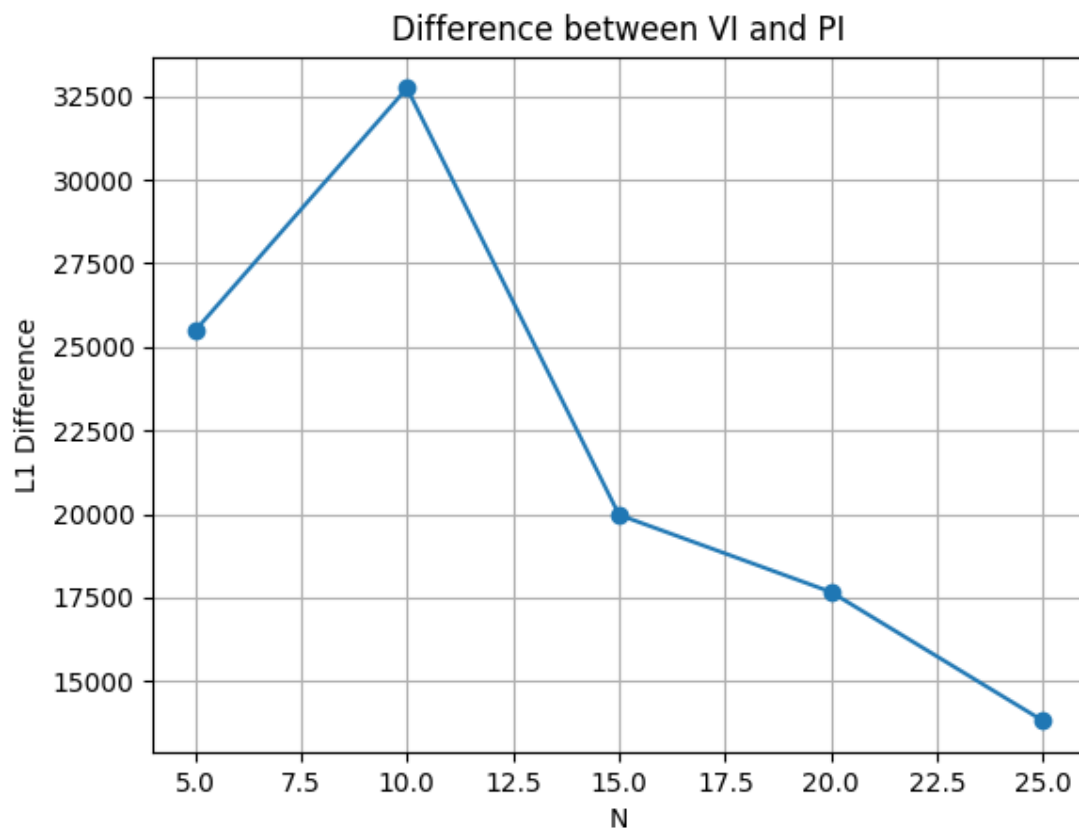
Policy Iteration consists of two main steps: policy evaluation and policy improvement. In the evaluation step, the value function for the current policy is estimated using an iterative method similar to Q1. In the improvement step, the policy is updated by selecting the action that maximizes the expected return for each state. These two steps are repeated until the policy becomes stable. As with Value Iteration, sampling is used to estimate expectations due to the large state space.

Q2(d). Comparing the L1 difference :

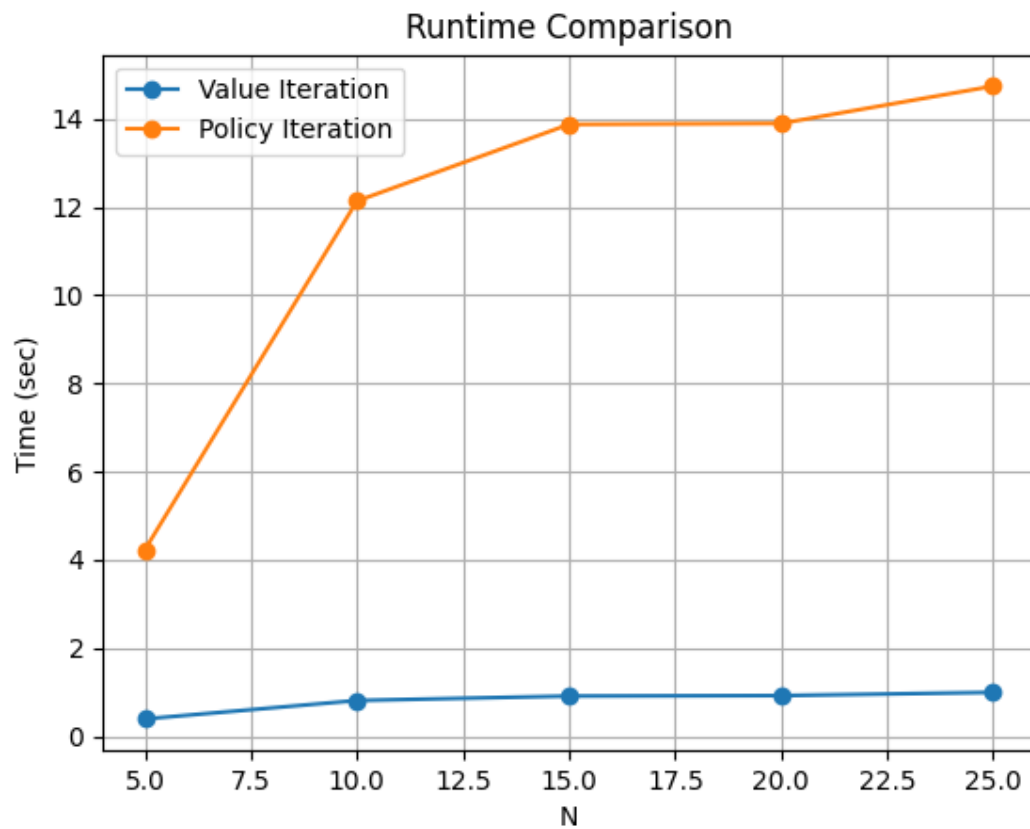
To compare the results of Value Iteration and Policy Iteration, the L1 difference between their Q-functions is computed. This measures how close the two methods are in terms of their estimated values. In theory, both methods should converge to the same optimal solution. However, due to sampling, limited iterations, and the use of a reduced state space, the difference is non-zero. The results show that the difference decreases for larger values of N, indicating that both methods produce similar approximations.

Results :

PLOT 1: L1 Difference between Value Iteration and Policy Iteration vs N



PLOT 2: Runtime Comparison vs N



The results demonstrate that both Value Iteration and Policy Iteration are effective methods for solving the problem, even under approximation. Value Iteration is faster, while Policy Iteration is more structured but slower. The use of sampling and state reduction enables the algorithms to scale to larger grid sizes while still capturing meaningful trends in both value estimation and runtime.